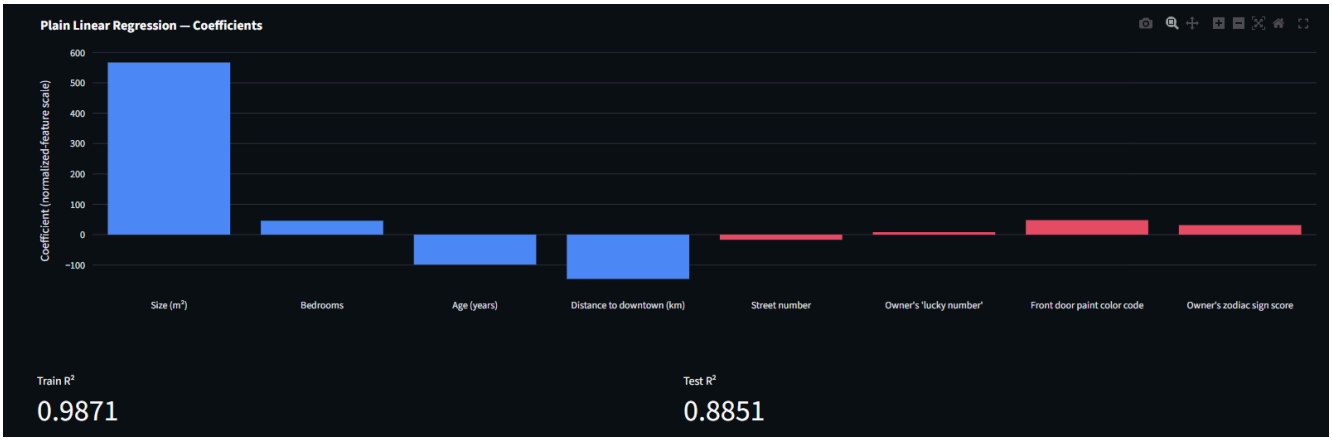
Activity 13 — Mérida Homes: Taming Overfitting with Regularization

Labeled screenshots · Session 23 · Diego Kamil Olea Ruiz · Artificial Intelligence

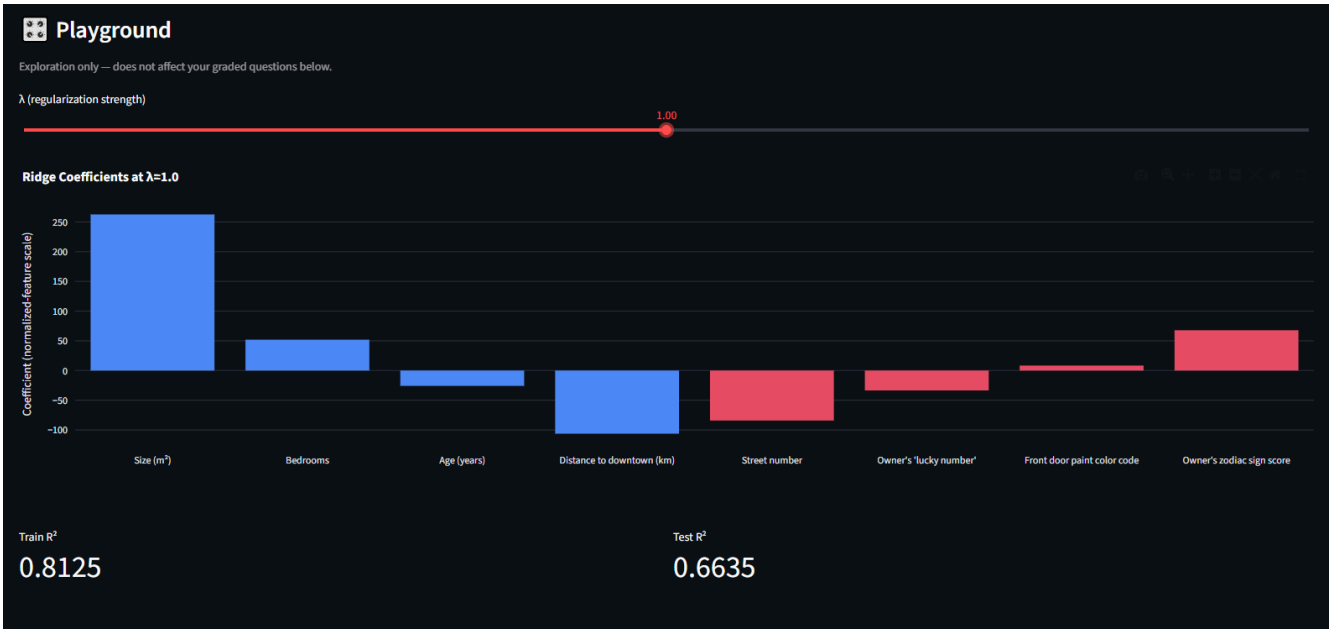
A — The Overfitting Problem: plain model

Supports question 1. This is the plain model with no regularization. Train R^2 is 0.9871 and test R^2 is 0.8851, so there is a gap of about 0.102. Basically it fits the 20 training houses really well but does worse on new ones, and the red noise features still get real weight.



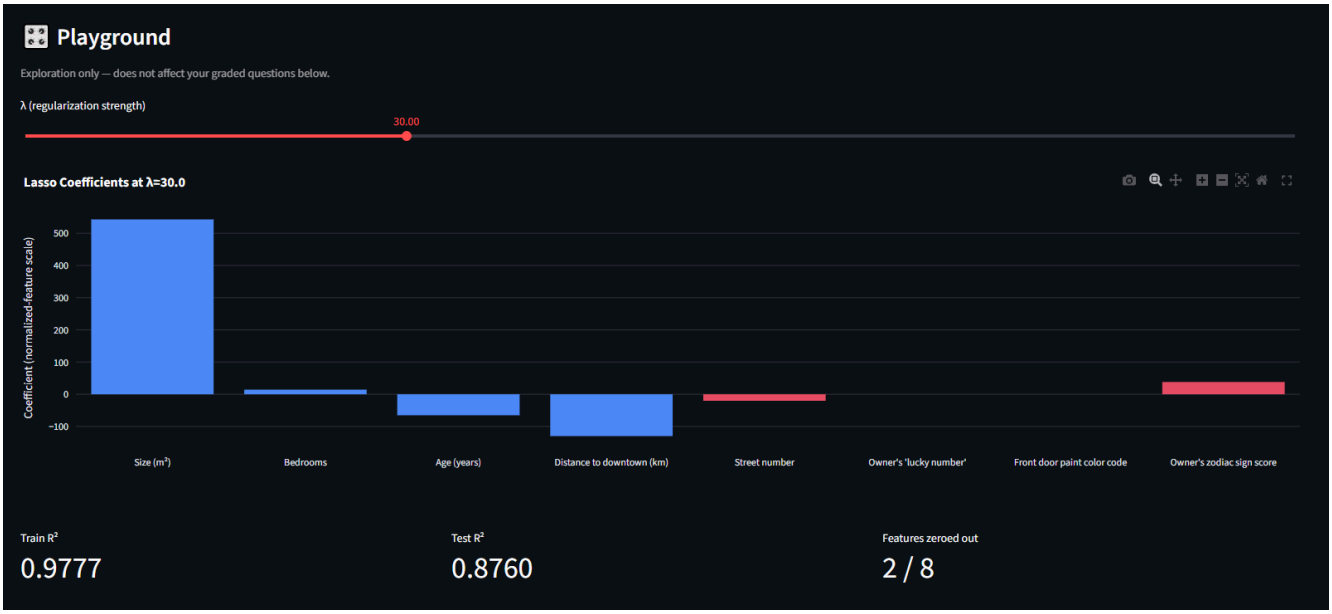
B — Ridge playground at a large λ

Supports questions 6 and 7. I moved λ to 1.0 and every bar got smaller, but none of them hit zero, not even the red ones. Test R^2 dropped to 0.6635, way worse than the plain model, so a bigger λ is not always better.



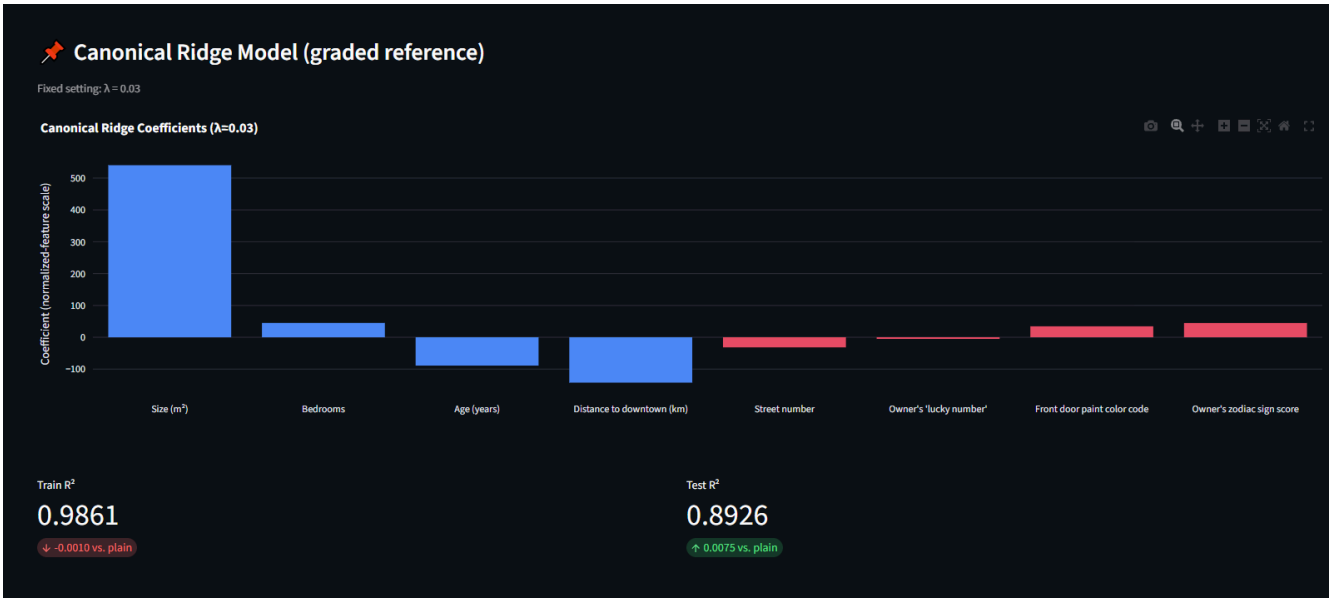
C — Lasso playground with features zeroed out

Supports question 6. At λ = 30 Lasso zeroed out 2 of the 8 features, the lucky number and the paint color code, while the 4 real features stayed nonzero.



D — Canonical Ridge Model (λ = 0.03)

Supports questions 2 and 3. The graded Ridge model gets train R^2 0.9861 and test R^2 0.8926. It loses a tiny bit on training but gains on test compared to the plain model, so it generalizes a little better.



E — Canonical Lasso Model (λ = 30)

Supports questions 4 and 5. The graded Lasso model gets train R^2 0.9777 and test R^2 0.8760, with 2 of 8 features zeroed out: the owner's lucky number and the front door paint color code.

